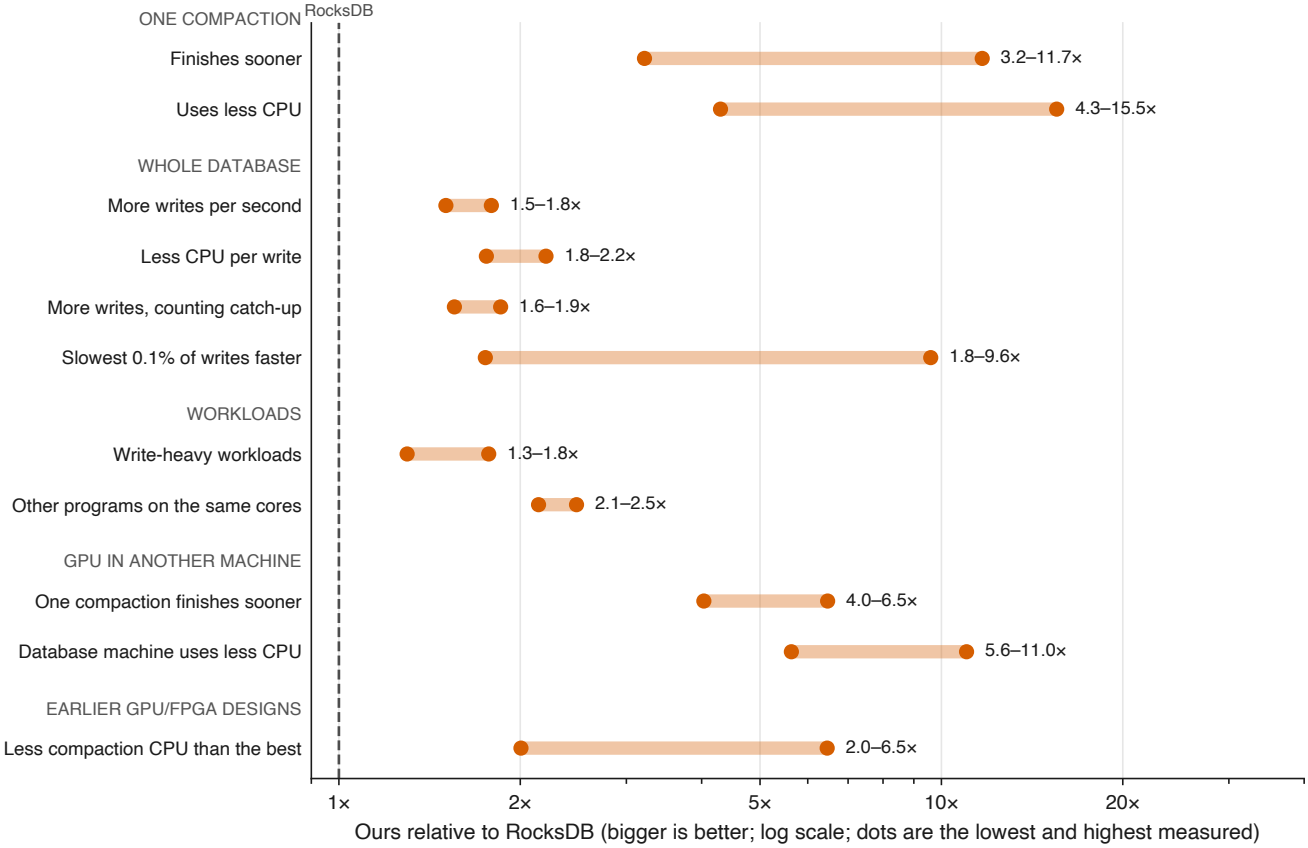
RocksDB
ONE COMPACTION
Finishes sooner
3.2–11.7×
Uses less CPU
4.3–15.5×
WHOLE DATABASE
More writes per second
1.5–1.8×
Less CPU per write
1.8–2.2×
More writes, counting catch-up
1.6–1.9×
Slowest 0.1% of writes faster
1.8–9.6×
WORKLOADS
Write-heavy workloads
1.3–1.8×
Other programs on the same cores
2.1–2.5×
GPU IN ANOTHER MACHINE
One compaction finishes sooner
4.0–6.5×
Database machine uses less CPU
5.6–11.0×
EARLIER GPU/FPGA DESIGNS
Less compaction CPU than the best
2.0–6.5×
1×
2×
5×
10×
20×
Ours relative to RocksDB (bigger is better; log scale; dots are the lowest and highest measured)